Supplementary Supporting Documents

Application for MSc in Robotics

The Chinese University of Hong Kong

Applicant	Yongyi Lai
Programme	MSc in Robotics
Purpose	Supplementary materials for the online application

Contents of this PDF

1	**Personal Statement**	Programme-specific statement of purpose
2	**Academic Reference Letter**	Academic recommendation from Shenzhen Technology University
3	**Professional Reference Letter**	Employer recommendation from Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)

Document order

The materials are arranged in the following order for ease of review: personal statement, academic reference letter, and professional reference letter.

Personal Statement — MSc in Robotics, The Chinese University of Hong Kong

Over the past two years, I have worked on three robots whose control had to be validated on real hardware: a RoboMaster hero robot for which I implemented the gimbal, launcher and chassis control loops; a wheel-legged robot that stays upright while its legs change length; and a three-omni-wheel ballbot that keeps its balance while two arms move on top of it. These projects led me to a clear conclusion: I can implement a controller and make it run, but I now need a more rigorous understanding of why it works and how to design it systematically. That is why I am applying to the MSc in Robotics at The Chinese University of Hong Kong.

My undergraduate study at Shenzhen Technology University (Computer Science, ranked 6 of 156) gave me a solid base in programming and algorithms. As my robotics projects increasingly required a stronger understanding of control, I also took Principles of Automatic Control and Modern Control Theory. During my internship at the Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), I applied this preparation to a real balance-control problem. I was responsible for the balance control of a dual-arm ballbot. I built and deployed a cascaded PID controller, then modelled the system as a four-state inverted pendulum and used LQR to analyse how state weights trade stability against control effort. The model explained tuning behaviour and actuator saturation that hand tuning alone could not. Compensating arm-induced centre-of-mass shifts with URDF/Pinocchio gravity feedforward showed me how tightly upper-body motion and base balance are coupled. I also integrated the ROS state processing with CAN motor execution and added state initialisation, output limiting and fall-recovery safeguards, so that the loop could be tested repeatedly without damaging the robot. The chassis ultimately maintained dynamic balance while both arms moved along fixed trajectories.

I then extended this control-oriented approach through two laboratory projects, moving from balance control to locomotion and navigation. On a variable-leg-length wheel-legged robot I implemented gain-scheduled LQR and virtual model control with closed-chain leg kinematics, and tuned it on an STM32H7/FDCAN platform. On a car-like mobile robot I built the full chain from EKF/AMCL localisation through Hybrid-A* and TEB planning to MPC tracking, and ran it on the real vehicle. Internships at Future Power and Kaihong then extended my experience in two directions: VR teleoperation of a dual-arm robot, where constrained inverse kinematics keeps the arms tracking the operator without velocity spikes near singularities, and multi-robot platform software on OpenHarmony.

The MSc in Robotics at The Chinese University of Hong Kong is the most direct academic continuation of this preparation. Its foundation in robotic systems, experimental robotics and autonomous mobile robots corresponds to the path I have followed from balance control to navigation and manipulation. The programme's coverage of kinematics, dynamics, sensing, control, path planning, robot learning and embodied intelligence would allow me to connect the separate modules I have implemented into a more principled understanding of complete robotic systems. I am also interested in the hands-on and project-based training, because my strongest learning has come from taking an algorithm from a model to a real platform and analysing what changes in that transition.

My goal is to work as a robotics algorithm engineer in the Greater Bay Area, focusing on motion control, navigation and learning for mobile and legged robots. I would bring practical experience with real-time control, state estimation, planning and embedded execution, while using the programme to strengthen the theoretical and experimental foundations that I still need for advanced robot design.

Recommendation Letter — Academic Advisor

Date: 21 September 2026

Dear Members of the Admissions Committee,

I am writing to support Yongyi Lai's application to the MSc in Robotics at The Chinese University of Hong Kong.

I am Associate Dean in the School of Artificial Intelligence at Shenzhen Technology University, where Yongyi is a final-year student in Computer Science and Technology. I have known Yongyi since 2023, first as the instructor of University Computer Science A, and since then as the supervisor of two robotics projects in my laboratory. I have also followed Yongyi's competition and independent work closely, and this letter draws on all three.

Yongyi first came to my attention as a strong student who deliberately extended a Computer Science background into robotics. The academic record itself is straightforward: an average of 87.63 and a rank of 6th of 156 in the cohort. What stood out was that Yongyi had taken Principles of Automatic Control and Modern Control Theory in addition to the major curriculum, and then applied that preparation to real robot projects. This combination of computing ability and purposeful study in control theory was the basis on which Yongyi took on the two projects I describe below.

The first project, a wheel-legged robot with variable leg length, tested this preparation under changing mechanical geometry: the balance controller had to remain effective as the robot's leg configuration changed. Yongyi constructed the body-pitch and motion states from IMU, wheel-speed and joint feedback, implemented an LQR controller with gains scheduled by leg length, and added a virtual model controller that derived virtual leg length and posture from the closed-chain leg kinematics and mapped support force and pitch torque to the joints through the Jacobian. The controller was integrated and tuned on the physical robot, running on an STM32H7/FreeRTOS platform with FDCAN. The robot kept its body balanced and tracked velocity commands while its legs extended and retracted. The tuning process mattered to me as much as the result: when balance failed, Yongyi located the causes in state inconsistency, torque limits and ground-contact transitions and corrected each in turn, demonstrating a disciplined approach to diagnosing a physical system.

The second project expanded this work from locomotion to navigation. On a car-like mobile robot, Yongyi was responsible for the autonomy chain: EKF/AMCL localisation fusing IMU, wheel encoders and 2D LiDAR, including the handling of timing and coordinate-frame inconsistencies that cause pose

jumps; Hybrid-A* global planning under the vehicle's kinematic constraints; TEB local trajectory optimisation; and an MPC tracking controller on a car-like kinematic model. The system ran on the physical vehicle and achieved autonomous navigation from localisation through obstacle-aware planning to trajectory tracking. In my experience, it is unusual at undergraduate level for a student to take responsibility for work spanning localisation, planning and control and carry it through to a functioning vehicle.

I have also seen the same capabilities in work I did not supervise directly but followed closely. In the university RoboMaster team Yongyi was responsible for the embedded control of the hero robot, implementing the gimbal, launcher and chassis control loops on STM32F407/FreeRTOS over CAN; the robot competed in the 2025 national final, where the team won third prize, and the team also took second prize in the 3V3 league. Yongyi also designed and built, alone, an OpenHarmony smart gardening system and demonstrated it to me: it runs YOLOv5 inference on an RK3588 edge device, makes decisions through a LangGraph agent chain and controls irrigation on a Hi3861 device, a complete loop from perception to actuation.

Taken together, these four pieces of work show a student who can move from control theory to embedded implementation and back, who finishes systems rather than components, and who treats documentation and hardware safety as part of the design. They also show where project work stops. In both laboratory projects Yongyi arrived at the model by working it out for that particular robot, and at the gains by tuning on the machine; the general theory of control, planning and learning that would let the same student design rather than rediscover is not something our undergraduate curriculum provides. Yongyi recognises this and has chosen a taught master's for that reason, which I consider the right decision at the right time.

I recommend Yongyi Lai for admission without hesitation, and I would be glad to provide any further information.

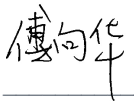
Yours sincerely,

Fu Xianghua

Associate Dean, School of Artificial Intelligence

Shenzhen Technology University

fuxianghua@sztu.edu.cn

Referee signature: 

SHENZHEN INSTITUTE
of ARTIFICIAL INTELLIGENCE AND ROBOTICS for SOCIETY
深圳市人工智能与机器人研究院

Recommendation Letter — AIRS Mentor

Date: 21 September 2026

Dear Members of the Admissions Committee,

I am writing to recommend Yongyi Lai for admission to the MSc in Robotics at The Chinese University of Hong Kong.

I am a Robot Development Engineer at the Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), where I supervised Yongyi directly as a robotics software intern from September 2025 to March 2026. During those six months Yongyi was responsible for the chassis balance control of a three-omni-wheel ballbot carrying a dual-arm upper body, a demanding control problem in our group. Under my supervision, Yongyi made steady progress toward this objective.

The task was to keep a dynamically unstable base balanced while two arms moved on top of it, which shifts the centre of mass continuously. Yongyi built the control state from IMU and wheel feedback, designed a cascaded controller in which a position outer loop generates bounded attitude references for an attitude inner loop, allocated the resulting actions to the three wheels through the wheel kinematics, and deployed and tuned the whole loop on the physical robot. To handle the arm disturbance, Yongyi estimated the centre-of-mass offset from the arm joint states using the robot's URDF model with Pinocchio and fed it into the balance controller as gravity feedforward, then verified the effect by comparing the balance response with compensation switched on and off under fixed arm trajectories. By the end of the internship the chassis maintained dynamic balance while both arms executed their trajectories, which was the objective we had set.

What stood out was Yongyi's ability to return to model-based analysis when empirical tuning could no longer fully explain the robot's behaviour. Rather than continuing to adjust gains without a model, Yongyi derived a four-state inverted-pendulum model of the ballbot, linearised it and used LQR to analyse how the weights on position, velocity, tilt and tilt rate trade stability against control effort. That analysis became our reference for interpreting the deployed controller's gains, identifying state anomalies and locating actuator saturation. It also demonstrated analytical ability beyond what I would normally expect from an undergraduate intern.

Yongyi was equally reliable on the engineering safeguards that make a hardware loop trustworthy: ROS state processing and CAN motor execution, state initialisation on re-enable, output limiting, and fall detection and recovery. These safeguards were implemented carefully and documented clearly, allowing the robot to be tested repeatedly under controlled conditions. Yongyi communicated progress and

problems concisely, asked precise questions, and worked independently once a direction was agreed.

Yongyi discussed with me the plan to take a taught master's in robotics or AI before working as an algorithm engineer. Given how much of this internship's progress came from Yongyi turning to model-based analysis when tuning ran out, I think that is the right next step. On the evidence of this internship, Yongyi has the technical ability, the hands-on discipline and the intellectual curiosity to benefit fully from your programme. I recommend Yongyi without reservation and would be glad to answer any questions.

Sincerely,

Chen Dansheng

Robot Development Engineer

Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)

chendansheng@cuhk.edu.cn

Referee signature: 